



Perceived Service Quality and Students' Satisfaction in a Higher Education Institution: A Structural Equation Modeling Approach

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ABSTRACT

Background: Increasing demands in terms of efficient operations, cost reduction, and improving productivity have emerged in the sugar industry due to increased competition and sustainable developments. The current pilot study suggests a framework to evaluate the impact that integration of AI, GIS, and IoT would bring on productivity of sugar mills.

Method: This research adopts a mixed frameworks approach to investigate the impact of integrated AI, GIS and IoT technologies on sugar mill productivity. Based on the simulated data that represents more than 20 sugar mills operating in Maharashtra, India, our findings suggest that adoption metrics concerning these technologies positively correlate with performance metrics such as productivity index, cost reduction, downtime reduction, and improvement in sugar recovery.

Result: AI adoption correlated most strongly with productivity index ($r = 0.264$), followed by integration of IoT technologies ($r = 0.119$) and GIS-based monitoring ($r = 0.042$). Random Forest regression provided R^2 value of 0.082. Among the most significant predictors, predictive maintenance score (14.8%) and scores of AI adoption and road quality index (13.5% each) were detected. The average productivity index was at 85.82, while average reductions in cost, downtime, and sugar recovery amounted to 14.12%, 31.92%, and 4.86% correspondingly.

Conclusion: In general, it could be noted that the proposed framework is applicable to various locations where sugar is produced. This study provides a methodology that can be replicated to explore the relationship between technology and productivity in the agro-industrial sector, while emphasizing the importance of predictive maintenance and AI decision-making systems to achieve business outcomes.

Key words: artificial intelligence, geographic information systems, internet of things, sugar industry, productivity analysis, predictive maintenance, digital transformation, operational efficiency

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INTRODUCTION

Education is an important service sector and a cornerstone of national development and advancement (Hill, 1995). In higher education, institutions are responsible not only for delivering academic knowledge but also for providing effective administrative, academic, and student support services. Therefore, the quality of these services is an important consideration in understanding students' overall educational experiences. Service quality is closely associated with customer satisfaction and loyalty in service-oriented organizations, and this relationship is increasingly relevant to higher education institutions (Kasiri et al., 2017). Customer satisfaction is particularly important because of its close connection with perceived service quality (Lupo, 2014).

Perceived service quality in higher education refers to students' overall evaluation of the quality of services provided by an educational institution. Students' perceptions may be influenced by teaching quality, institutional facilities, administrative support, and the responsiveness of the institution to their needs. Quality is also con-

sidered a strategic means of improving organizational performance and meeting established quality standards (DeShields et al., 2005). Service quality is commonly assessed through five dimensions: tangibility, reliability, responsiveness, assurance, and empathy, which collectively represent important aspects of students' experiences within the educational environment (Daniel and Berinyuy, 2010).

Understanding and improving perceived service quality is essential for creating a supportive learning environment and achieving meaningful educational outcomes (Becket and Brookes, 2006). Students' satisfaction may also vary according to their sociodemographic characteristics. Therefore, examining both sociodemographic factors and perceived service quality can provide a broader understanding of students' satisfaction. Therefore, this study aimed to examine the association between students' satisfaction and sociodemographic characteristics and to assess the relationship between perceived service quality and students' satisfaction.

METHODS

An analytical cross-sectional study was conducted among Bachelor's and Master's level students of Birendra Multiple Campus, Bharatpur-10, Chitwan, Nepal, from January to April 2024. The study aimed to examine the association between students' satisfaction and sociodemographic characteristics and to assess the relationship between perceived service quality and students' satisfaction. The study population comprised Bachelor's and Master's level students enrolled at Birendra Multiple Campus.

The sample size was calculated using the single population proportion formula:

$$n = \frac{Z^2 pq}{e^2} \quad (1)$$

where n is the required sample size, Z is the standard normal deviate corresponding to the desired confidence level, p is the estimated proportion of satisfaction, $q = 1 - p$, and e is the margin of error.

For a 95% confidence level, 50% satisfaction level, and 5% margin of error,

Where, $Z = 1.96$, $p = 0.50$, $q = 0.50$, $e = 0.05$

Therefore,

$$n = \frac{(1.96)^2(0.50)(0.50)}{(0.05)^2} \quad (2)$$

$$n = \frac{3.8416 \times 0.25}{0.0025} = 384.16 \approx 384 \quad (3)$$

Thus, calculated sample size was 384 students. Participants were selected using a non-probability purposive sampling technique.

Data were collected using a structured questionnaire. The questionnaire included sections on sociodemographic characteristics, perceived service quality, and students' satisfaction. Sociodemographic characteristics included age, gender, ethnicity, religion, place of residence, family monthly income, faculty, and academic year. Perceived service quality was assessed across five dimensions: tangibility, reliability, responsiveness, assurance, and empathy. Tangibility referred to the physical and observable aspects of institutional services and was measured using five items (T1–T5); reliability assessed the consistency, dependability, and accuracy of service delivery using five items (R1–R5); responsiveness assessed the willingness and promptness of service providers in addressing students' needs using four items (Re1–Re4); assurance assessed competence, courtesy, credibility, trust, and confidence using four items (As1–As4); and empathy assessed the extent to which service providers understand and respond to students' individual needs using five items (Em1–Em5). Students' satisfaction was assessed using nine items (Sa1–Sa9). All service-quality and satisfaction items were measured using a five-point Likert scale ranging from strongly agree (5) to strongly disagree (1).

The collected data were checked for completeness, coded, and analyzed using SPSS, R, and Jamovi software. Descriptive statistics were used to summarize the characteristics of the participants and study variables. Frequencies and percentages were calculated for categorical variables, while means and standard deviations were

calculated for scale-level variables. The Chi-square test was used to examine the association between students' satisfaction and selected sociodemographic characteristics. Structural Equation Modeling (SEM) was employed to examine the relationships between the dimensions of perceived service quality and students' satisfaction.

Structural Equation Modeling

Structural Equation Modeling (SEM) was used to examine the hypothesized relationship between perceived service quality and students' satisfaction while accounting for measurement error in the latent constructs. The analysis consisted of two related components: the measurement model, which assessed whether the observed questionnaire items adequately represented their respective latent constructs, and the structural model, which evaluated the hypothesized relationships among the constructs (Pradhan, 2017). The measurement model was expressed as

$$\mathbf{x} = \mathbf{\Lambda}_x \boldsymbol{\xi} + \boldsymbol{\delta} \quad (4)$$

where $\mathbf{x}$ and $\mathbf{y}$ denote observed indicators, $\boldsymbol{\xi}$ and $\boldsymbol{\eta}$ represent latent constructs, $\mathbf{\Lambda}_x$ and $\mathbf{\Lambda}_y$ are factor-loading matrices, and $\boldsymbol{\delta}$ and $\boldsymbol{\epsilon}$ represent measurement errors. The structural relationship was specified as

$$\eta = \beta \xi + \zeta \quad (5)$$

where η represents students' satisfaction, ξ represents perceived service quality, β is the structural path coefficient representing the effect of perceived service quality on students' satisfaction, and ζ is the structural disturbance term.

The adequacy of the measurement model was assessed using standardized factor loadings, Composite Reliability (CR), and Average Variance Extracted (AVE), which are commonly used to evaluate the reliability and validity of latent constructs in confirmatory measurement models (Pradhan, 2017; Pradhan et al., 2023), with

$$CR = \frac{\left(\sum_{i=1}^k \lambda_i\right)^2}{\left(\sum_{i=1}^k \lambda_i\right)^2 + \sum_{i=1}^k \theta_i} \quad (6)$$

and

$$AVE = \frac{\sum_{i=1}^k \lambda_i^2}{\sum_{i=1}^k \lambda_i^2 + \sum_{i=1}^k \theta_i} \quad (7)$$

where λ_i denotes the standardized factor loading and θ_i denotes the corresponding measurement-error variance.

Overall Model Fit and Structural Path Assessment

Overall model fit was evaluated by comparing the observed covariance matrix $\mathbf{S}$ with the model-implied covariance matrix $\boldsymbol{\Sigma}(\boldsymbol{\theta})$. The chi-square goodness-of-fit statistic was used to test the null hypothesis that the model-implied covariance matrix does not differ from the observed covariance matrix:

$$\chi^2 = (N - 1)F_{\min} \quad (8)$$

with

$$F_{\min} = F[\mathbf{S}, \boldsymbol{\Sigma}(\boldsymbol{\theta})] \quad (9)$$

where N is the sample size and $F_{\min}$ is the minimum value of the fitting function. The chi-square statistic was interpreted together with its degrees of freedom because it is sensitive to sample size. The relative chi-square was calculated as

$$\frac{\chi^2}{df} \quad (10)$$

with smaller values indicating less discrepancy between the hypothesized and observed covariance structures.

Incremental model fit was assessed using the Comparative Fit Index (CFI) and Tucker–Lewis Index (TLI). CFI compares the proposed model with an independence or baseline model and can be expressed as

$$CFI = 1 - \frac{\max(\chi_m^2 - df_m, 0)}{\max(\chi_b^2 - df_b, 0)} \quad (11)$$

where the subscripts m and b denote the proposed and baseline models, respectively. TLI evaluates improvement in fit while accounting for model complexity and can be expressed as

$$TLI = \frac{\left(\frac{\chi_b^2}{df_b}\right) - \left(\frac{\chi_m^2}{df_m}\right)}{\left(\frac{\chi_b^2}{df_b}\right) - 1} \quad (12)$$

Approximate fit was assessed using the Root Mean Square Error of Approximation (RMSEA),

$$RMSEA = \sqrt{\frac{\max(\chi^2 - df, 0)}{df(N-1)}} \quad (13)$$

which reflects the degree of model misfit per degree of freedom. The Standardized Root Mean Square Residual (SRMR) was also reported as a measure of the average standardized difference between the observed and model-implied correlations/covariances. Values of CFI and TLI closer to 1 and lower values of RMSEA and SRMR indicate better fit. As commonly used guidelines, CFI and TLI values around 0.95 or higher and RMSEA and SRMR values around 0.06 and 0.08 or lower, respectively, indicate relatively good fit, although these values were not treated as rigid universal cutoffs.

The hypothesized relationship between perceived service quality and students' satisfaction was assessed using the estimated structural path coefficient. Its standard error was used to obtain the test statistic,

$$z = \frac{\hat{\beta}}{SE(\hat{\beta})} \quad (14)$$

and the corresponding two-sided p -value. A $(1 - \alpha) \times 100\%$ confidence interval was calculated as

$$\hat{\beta} \pm z_{\alpha/2} SE(\hat{\beta}) \quad (15)$$

The relationship was considered statistically significant at the 5% significance level if $p < 0.05$, or equivalently,

if the 95% confidence interval for the path coefficient did not include zero. The magnitude and direction of β were used to assess the strength and direction of the hypothesized relationship.

RESULTS

Among the 392 students, the mean age was 22.12 ± 4.34 years, with the largest proportion aged 20–25 years (46.9%), followed by those aged < 20 years (42.6%). Females constituted 55.4% of the participants, while 74.5% belonged to the Brahmin/Chhetri ethnic group.

Most participants were Hindu (93.4%), while 5.4% were Buddhist and the remaining participants followed other religions. In terms of marital status, the majority were unmarried (87.8%), whereas 11.7% were married and 0.5% reported other marital statuses. This distribution reflects the predominantly young student population included in the study.

Regarding occupation, half of the participants (50.8%) were associated with agriculture, followed by those abroad (20.7%) and those engaged in service (19.9%). The remaining 8.7% were classified under other occupations. Participants were almost equally distributed by place of residence, with 50.3% living in rural areas and 49.7% in urban areas. Regarding monthly family income, the largest proportion (37.5%) reported an income of 50,000–1 lakh, while 33.4% reported less than 50,000 and 29.1% reported 1 lakh or more.

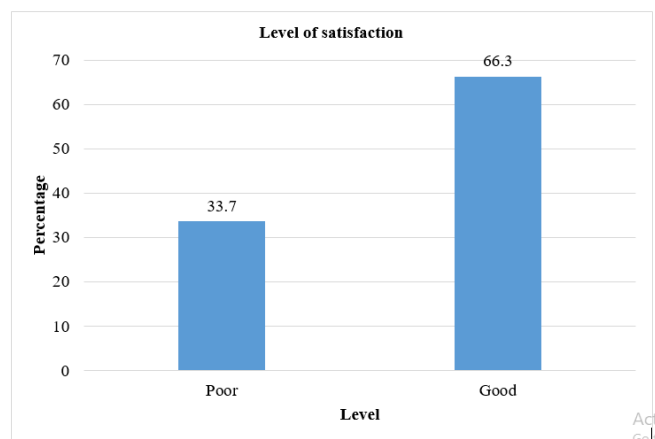
By educational stream, Science students constituted the largest proportion (35.97%), followed by Arts (28.83%), Management (28.06%), and Education (7.14%). Most participants were pursuing Bachelor's degrees (79.34%), whereas 20.66% were enrolled in Master's degree programs. Among the individual programs, MA had the highest representation (13.01%), followed by BBA (11.48%), BBS (8.93%), and BIT (8.16%). The remaining participants were distributed across BSc Physical Group, BSc Biology Group, BSc Microbiology, BSc Geology, BSc CSIT, B.Ed., BCA, and MBS programs, as presented in Table ??.

Overall, 66.3% of the students reported satisfaction with the various aspects of their academic environment. The estimated proportion of satisfied students was 66.3%, with a 95% confidence interval ranging from 61.41% to 71.00%. This suggests that approximately two-thirds of the students were satisfied with their academic environment, while a considerable proportion remained dissatisfied (Figure 1).

Table 2 presents the association between students' level of satisfaction and selected sociodemographic and educational characteristics. A statistically significant association was observed between age and satisfaction ($\chi^2 = 17.65$, $p = 0.001$). The proportion of students reporting good satisfaction increased with age, from 61.7% among those aged below 20 years to 64.1% among those aged 20–25 years and 90.5% among those aged 25–30 years. All participants aged 30–35 years and above 35 years re-

Table 1. Sociodemographic characteristics of students
($n = 392$)

Variables	Frequency	Percentage (%)
Age (years)		
< 20	167	42.6
20–25	184	46.9
25–30	21	5.4
30–35	10	2.6
≥ 35	10	2.6
Mean $\pm$ SD	22.12 $\pm$ 4.34	
Gender		
Male	175	44.6
Female	217	55.4
Ethnicity		
Brahmin/Chhetri	292	74.5
Dalit/Janajati	50	12.8
Terai/Madhesi	20	5.1
Others	30	7.7
Religion		
Hindu	366	93.4
Buddhist	21	5.4
Christian	2	0.5
Others	3	0.8
Marital status		
Unmarried	344	87.8
Married	46	11.7
Others	2	0.5
Occupation		
Agriculture	199	50.8
Service	78	19.9
Abroad	81	20.7
Others	34	8.7
Place of residence		
Urban	195	49.7
Rural	197	50.3
Monthly income		
< 50,000	131	33.4
50,000–1 lakh	147	37.5
≥ 1 lakh	114	29.1
Stream of education		
Science	141	35.97
Management	110	28.06
Education	28	7.14
Arts	113	28.83
Level of education		
Bachelor degree	311	79.34
Master degree	81	20.66
Faculty/Program		
BSc Physical group	31	7.91
BSc Biology group	25	6.38
BSc Microbiology	28	7.14
BSc Geology	30	7.65
BSc CSIT	27	6.89
B.Ed.	28	7.14
BIT	32	8.16
BCA	30	7.65
BBA	45	11.48
BBS	35	8.93
MBS	30	7.65
MA	51	13.01

**Figure 1.** Overall level of satisfaction.

ported good satisfaction. However, these older age groups had relatively small numbers of participants, and therefore the observed differences should be interpreted with caution.

No statistically significant association was found between gender and satisfaction ($\chi^2 = 0.172$, $p = 0.678$). Good satisfaction was reported by 67.4% of male students and 65.4% of female students, indicating only a small difference between the two groups. Similarly, satisfaction did not differ significantly by ethnicity ($\chi^2 = 2.141$, $p = 0.710$). The proportion reporting good satisfaction ranged from 66.1% among Brahmin/Chhetri students to 70.0% among both Dalit/Janajati and Terai/Madhesi students.

Religion was also not significantly associated with satisfaction ($\chi^2 = 1.200$, $p = 0.753$). Good satisfaction was reported by 66.4% of Hindu students and 61.9% of Buddhist students. However, the very small numbers in the Christian and other religious groups should be considered when interpreting these results. Likewise, marital status was not significantly associated with satisfaction ($\chi^2 = 3.324$, $p = 0.190$). Although married students reported a higher proportion of good satisfaction than unmarried students (76.1% vs. 64.8%), the difference was not statistically significant.

Occupation was not significantly associated with satisfaction ($\chi^2 = 3.312$, $p = 0.346$). Good satisfaction was highest among participants associated with other occupations (69.1%), followed by agriculture (68.8%), service (61.5%), and those abroad (55.9%). Similarly, place of residence did not show a statistically significant association with satisfaction ($\chi^2 = 3.175$, $p = 0.075$). Rural students reported a higher proportion of good satisfaction than urban students (70.6% vs. 62.1%); however, this difference did not reach the conventional level of statistical significance.

Monthly family income was also not significantly associated with satisfaction ($\chi^2 = 0.880$, $p = 0.640$). The proportion of students reporting good satisfaction was 66.4% among those with a monthly income below 50,000, 68.7% among those with an income of 50,000–1 lakh, and 63.2% among those with an income of 1 lakh or more. Thus, satisfaction levels were relatively similar across the

Table 2. Association of satisfaction with sociodemographic variables

Variables	Level of satisfaction		Chi-square	p-value
	Poor	Good		
Age (years)				
< 20	64 (38.3)	103 (61.7)	17.65	0.001*
20–25	66 (35.9)	118 (64.1)		
25–30	2 (9.5)	19 (90.5)		
30–35	0 (0.0)	10 (100)		
> 35	0 (0.0)	10 (100)		
Gender				
Male	57 (32.6)	118 (67.4)	0.172	0.678
Female	75 (34.6)	142 (65.4)		
Ethnicity				
Brahmin/Chhetri	99 (33.9)	193 (66.1)	2.141	0.710
Dalit/Janajati	15 (30.0)	35 (70.0)		
Terai/Madhesi	6 (30.0)	14 (70.0)		
Others	5 (31.3)	11 (68.8)		
Religion				
Hindu	123 (33.6)	243 (66.4)	1.200	0.753*
Buddhist	8 (38.1)	13 (61.9)		
Christian	0 (0.0)	2 (100)		
Others	1 (33.3)	2 (66.7)		
Marital status				
Unmarried	121 (35.2)	223 (64.8)	3.324	0.190
Married	11 (23.9)	35 (76.1)		
Others	0 (0.0)	2 (100)		
Occupation				
Agriculture	62 (31.2)	137 (68.8)	3.312	0.346
Service	30 (38.5)	48 (61.5)		
Abroad	15 (44.1)	19 (55.9)		
Others	25 (30.9)	56 (69.1)		
Place of residence				
Urban	74 (37.9)	121 (62.1)	3.175	0.075
Rural	58 (29.4)	139 (70.6)		
Monthly income (Rs.)				
< 50,000	44 (33.6)	87 (66.4)	0.880	0.640
50,000–1 lakh	46 (31.3)	101 (68.7)		
≥ 1 lakh	42 (36.8)	72 (63.2)		
Stream of education				
Science	56 (35.0)	104 (65.0)	19.63	< 0.0001
Management	62 (42.5)	84 (57.5)		
Education	10 (25.5)	29 (74.4)		
Arts	4 (8.5)	43 (91.5)		
Level of education				
Bachelor degree	128 (36.8)	220 (63.2)	13.41	< 0.001
Master degree	4 (9.1)	40 (90.9)		

*Likelihood ratio test value.

income categories.

In contrast, educational characteristics showed significant associations with satisfaction. A statistically significant association was observed between educational stream and satisfaction ($\chi^2 = 19.63$, $p < 0.0001$). Good satisfaction was highest among Arts students (91.5%), followed by Education (74.4%), Science (65.0%), and Management students (57.5%). This indicates considerable variation in satisfaction across different educational streams.

Similarly, level of education was significantly associated with satisfaction ($\chi^2 = 13.41$, $p < 0.001$). A substantially

higher proportion of Master's-level students reported good satisfaction (90.9%) compared with Bachelor's-level students (63.2%). This difference suggests that postgraduate students in the study population had more favorable perceptions of their academic environment than undergraduate students.

Overall, the findings demonstrate that age, educational stream, and level of education were significantly associated with students' satisfaction. In contrast, gender, ethnicity, religion, marital status, occupation, place of residence, and monthly family income did not show statistically significant associations with satisfaction. Among

the significant factors, educational stream and level of education showed particularly notable differences in the proportion of students reporting good satisfaction.

Structural Equation Modeling

The Structural Equation Modeling (SEM) results indicated that the proposed model demonstrated a good overall fit to the observed data. The SRMR value of 0.046 and RMSEA value of 0.047 indicated a close fit between the hypothesized model and the observed data. The χ^2/df ratio of 3.23 was within the generally acceptable range, suggesting an adequate balance between model fit and complexity. Furthermore, the CFI (0.963), TLI (0.950), NFI (0.962), and GFI (0.943) exceeded the commonly accepted threshold of 0.90, indicating satisfactory to good model fit. The AGFI value of 0.931 further supported the adequacy of the model after accounting for model complexity (Table 3).

The structural model demonstrated that all five dimensions of perceived service quality were positively and significantly associated with students' satisfaction. Tangibility showed a significant positive association with satisfaction ($\beta = 0.324$, 95% CI: 0.1017–0.546, $p = 0.006$), indicating that better physical facilities and observable aspects of institutional services were associated with higher student satisfaction. Similarly, reliability was positively associated with satisfaction ($\beta = 0.295$, 95% CI: 0.0966–0.4941, $p = 0.006$), suggesting that consistent and dependable service delivery contributed to greater satisfaction. Responsiveness also demonstrated a significant positive relationship with students' satisfaction ($\beta = 0.446$, 95% CI: 0.335–0.506, $p = 0.040$). Assurance was positively associated with satisfaction ($\beta = 0.249$, 95% CI: –0.0994–0.597, $p = 0.004$), while empathy showed the strongest positive association among the five dimensions ($\beta = 0.531$, 95% CI: 0.1904–0.8708, $p = 0.003$) (Table 4).

The value of adjusted R^2 in SEM is 0.769, which indicates that 76.9% of the variation in satisfaction is explained by the service quality domains included in the model, while the remaining variation in the dependent variable is due to other factors that are not included in the model. This indicates that the model has relatively strong explanatory power (Table 5).

The reliability and validity analysis demonstrated that all constructs exhibited satisfactory measurement properties, consistent with the evaluation of latent constructs through confirmatory factor analysis and related reliability and validity measures (Pradhan, 2017; Pradhan et al., 2023). Cronbach's alpha values ranged from 0.70 to 0.89 across Tangibility, Reliability, Responsiveness, Assurance, Empathy, and Satisfaction, with all values meeting the recommended threshold of 0.70, indicating acceptable internal consistency. Similarly, all Composite Reliability (CR) values exceeded 0.70, further supporting the reliability of the constructs. The Average Variance Extracted (AVE) values were above 0.50 for all constructs, providing evidence of satisfactory convergent validity and indicating that the constructs adequately explained the variance in their respective indicators (Table 6).

The discriminant validity analysis showed that the diagonal values, representing the square roots of AVE,

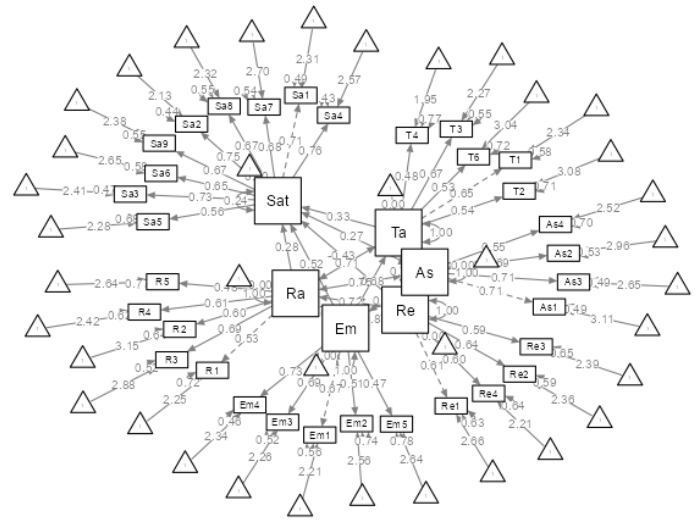


Figure 2. Structural equation model

ranged from 0.714 to 0.762 across the six constructs. These values were greater than the corresponding inter-construct correlations, providing evidence that the constructs were sufficiently distinct from one another. The correlations among the constructs were positive and statistically significant at the 0.01 level, ranging from 0.532 to 0.673, with the strongest association observed between Empathy and Satisfaction ($r = 0.673$). The results support adequate discriminant validity, indicating that Tangibility, Reliability, Responsiveness, Assurance, Empathy, and Satisfaction represent related but distinct constructs in the measurement model (Table 7).

The SEM results showed that all observed items had positive factor loadings on their respective latent constructs, indicating that the items adequately represented the underlying domains. For Tangibility, the loadings ranged from 0.461 to 0.674; for Reliability, from 0.483 to 0.692; and for Responsiveness, from 0.589 to 0.637. The Assurance items showed loadings ranging from 0.546 to 0.713, while Empathy loadings ranged from 0.468 to 0.734. The Satisfaction items demonstrated loadings between 0.562 and 0.758, with Sa4 showing the highest loading (0.758) (Figure 2).

DISCUSSION

The participants were predominantly young students, with a mean age of 22.12 ± 4.34 years, and represented different educational streams and levels. The present study found that 66.3% of students were satisfied with various aspects of the academic environment (95% CI: 61.41%–71.00%), with the median score being 26 (out of 45) for satisfaction. In comparison, Singh and Dhakal (2025) reported an overall median score of 147.0 out of 200, with 89.9% of students reporting high satisfaction. This finding was higher than the level reported by Shrestha et al. (2019), who reported an overall score of 131.25. Similarly, Mirghani and Elnour (2017) found that 74.5% of students viewed the academic environment positively. However, the present finding was substantially higher than that reported by Kasemy et al. (2022), who

Table 3. Model fitting criteria in structural equation modeling

Indicator	Calculated value	Recommended value
SRMR	0.046	Below 0.08 or 0.10
RMSEA	0.047	< 0.05
Chi-Square/df	3.23	Between 2 and 5
CFI	0.963	> 0.90
TLI	0.950	> 0.90
NFI	0.962	> 0.90
GFI	0.943	> 0.90
AGFI	0.931	> 0.90

Table 4. Structural model equation

Independent variables	β	95% Confidence Interval		Estimate	SE	z value	p-value
		Lower	Upper				
Tangibility	0.324	0.1017	0.5460	0.387	0.141	2.75	0.006
Reliability	0.295	0.0966	0.4941	0.420	0.152	2.76	0.006
Responsiveness	0.446	0.3350	0.5060	0.589	0.336	1.75	0.040
Assurance	0.249	-0.0994	0.5970	0.304	0.219	2.39	0.004
Empathy	0.531	0.1904	0.8708	0.618	0.211	2.93	0.003

Table 5. Adjusted R-squared (R^2)

Dependent variable	Independent variables	R^2
Satisfaction	Tangibility, Reliability, Responsiveness, Assurance, Empathy	0.769

found an overall score indicating a low level of satisfaction. In contrast, [Abusaad et al. \(2015\)](#) reported a neutral level of satisfaction. These differences may reflect variations in institutional settings, student characteristics, and educational environments.

The findings showed that age, stream of education, and level of education were significantly associated with students' satisfaction, suggesting that students' experiences and expectations may vary according to their demographic and academic characteristics. In the present study, quality of teaching was the highest-rated aspect of the academic environment ([Singh and Dhakal, 2025](#)), whereas ? identified learning facilities and resources as the highest-rated domain. Furthermore, students' satisfaction was significantly associated with academic year ($p = 0.038$), which is consistent with ?, who also reported a significant association ($p = 0.02$).

A key finding of the study was the significant positive relationship between perceived service quality and students' satisfaction. All five dimensions of service quality, namely tangibility, reliability, responsiveness, assurance, and empathy, were positively associated with students' satisfaction. Among these dimensions, empathy showed the strongest association, followed by responsiveness, tangibility, reliability, and assurance. This finding suggests that students' satisfaction is not determined solely by physical facilities or academic resources but is also closely related to how effectively and respectfully institutional services respond to their needs. In particular, attention to students' individual concerns and timely re-

sponses may play an important role in creating a more satisfying academic experience.

The structural model further indicated that perceived service quality explained 76.9% of the variance in students' satisfaction, demonstrating a substantial relationship between the quality of institutional services and students' overall satisfaction. This finding emphasizes the importance of considering multiple dimensions of service quality when evaluating the student experience. Improvements in institutional facilities, consistency of services, responsiveness of staff, professional support, and individualized attention may therefore be important areas for institutions seeking to enhance student satisfaction.

The significant associations observed for age, educational stream, and level of education also suggest that a one-size-fits-all approach to student services may not adequately address the needs of all students. Different groups of students may have different academic expectations, support requirements, and experiences within the institution. Institutional authorities should therefore consider differences across student groups when planning academic and support services.

CONCLUSIONS

The study concludes that more than half of the students were satisfied with the academic environment, although satisfaction varied across different student characteristics. Age, stream of education, and level of education were associated with students' satisfaction. Perceived service quality was also positively associated with satisfaction

Table 6. Internal and convergent reliabilities

Latent	Observed	Cronbach's α	Loading	z	p -value	Error	AVE	CR
Tangibility	T1	0.70	0.647	0	–	0.0871	0.53	0.71
	T2	–	0.536	8.76	< 0.001	0.0871	–	–
	T3	–	0.674	10.47	< 0.001	0.0901	–	–
	T4	–	0.481	7.99	< 0.001	0.0962	–	–
	T5	–	0.530	8.67	< 0.001	0.0794	–	–
Reliability	R1	0.73	0.526	0	–	0.1227	0.51	0.72
	R2	–	0.604	8.39	< 0.001	0.1227	–	–
	R3	–	0.692	9.04	< 0.001	0.1331	–	–
	R4	–	0.613	8.46	< 0.001	0.1204	–	–
	R5	–	0.483	7.26	< 0.001	0.1216	–	–
Responsiveness	Re1	0.71	0.609	0	–	0.0989	0.57	0.74
	Re2	–	0.637	10.05	< 0.001	0.0989	–	–
	Re3	–	0.589	9.47	< 0.001	0.1013	–	–
	Re4	–	0.600	9.61	< 0.001	0.1074	–	–
Assurance	As1	0.75	0.712	0	–	0.0804	0.54	0.76
	As2	–	0.686	12.37	< 0.001	0.0804	–	–
	As3	–	0.713	12.83	< 0.001	0.0845	–	–
	As4	–	0.546	9.94	< 0.001	0.0849	–	–
Empathy	Em1	0.74	0.665	0	–	0.0881	0.58	0.75
	Em2	–	0.509	8.98	< 0.001	0.0881	–	–
	Em3	–	0.690	11.75	< 0.001	0.0867	–	–
	Em4	–	0.734	12.35	< 0.001	0.0876	–	–
	Em5	–	0.468	8.32	< 0.001	0.0902	–	–
Satisfaction	Sa1	0.89	0.711	0	–	0.0746	0.53	0.89
	Sa2	–	0.747	14.04	< 0.001	0.0746	–	–
	Sa3	–	0.728	13.70	< 0.001	0.0725	–	–
	Sa4	–	0.758	14.25	< 0.001	0.0681	–	–
	Sa5	–	0.562	10.60	< 0.001	0.0719	–	–
	Sa6	–	0.652	12.28	< 0.001	0.0698	–	–
	Sa7	–	0.678	12.76	< 0.001	0.0672	–	–
	Sa8	–	0.670	12.61	< 0.001	0.0680	–	–
	Sa9	–	0.670	12.61	< 0.001	0.0661	–	–

Table 7. Discriminant validity using the Fornell–Larcker criterion

Domains	Tan.	Rel.	Resp.	Assu.	Emp.	Satis.
Tangibility	0.728	0.580**	0.538**	0.532**	0.532**	0.594**
Reliability	0.580**	0.714	0.584**	0.569**	0.647**	0.647**
Responsiveness	0.538**	0.584**	0.755	0.560**	0.636**	0.659**
Assurance	0.532**	0.569**	0.560**	0.735	0.644**	0.670**
Empathy	0.532**	0.647**	0.636**	0.644**	0.762	0.673**
Satisfaction	0.594**	0.647**	0.659**	0.670**	0.673**	0.728

Note: Diagonal elements in bold represent the square roots of the Average Variance Extracted (AVE). Off-diagonal elements represent Pearson correlations between the constructs. ** Significant at the 0.01 level (2-tailed).

across all five dimensions, namely tangibility, reliability, responsiveness, assurance, and empathy. The findings indicate that students' satisfaction depends not only on physical facilities and academic resources but also on the reliability and responsiveness of institutional services, the competence and professionalism of service providers, and the extent to which students' individual needs are understood and addressed. Therefore, higher education institutions should regularly assess students' perceptions of service quality and prioritize improvements in these areas to create a supportive, responsive, and student-centered academic environment.

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